

John Ott

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EDUCATION

Appalachian State University — Bachelor of Science in Computer Science — May 2025

Relevant Coursework: Data Structures, Software Engineering, Computer Systems, Databases, Programming Languages

WORK EXPERIENCE

Freelance Developer – AI-Driven Discord Bot – Python, OpenAI API

- Engineered a discord bot in Python that tracks and responds to a specific user across channels using context-aware AI-generated messages.
- Implemented event-driven architecture and message parsing to enable real-time interaction and behavior targeting.

Spread Goodness – Software Developer Intern – React.js, Firebase

Sep. 2025 - Present

- Developing and enhancing features for a mission-driven kindness-sharing platform using React.js and Firebase (Firestore, Auth).
- Collaborating with the front-end team (React.js) to integrate backend services, focusing on performance, scalability, and ease of use.

PROJECTS

GameBoy Emulator (“SimpleBoy” In-Progress) – Zig

- Built a functional GameBoy CPU in Zig with 8-bit registers, flag logic, and accurate cycle timing.
- Engineered a Memory Management Unit supporting ROM banking (MBC1, MBC2, MBC3) and boot ROM loading.
- Integrated instruction decoding, execution, and state machine logic for an authentic emulation experience.
- Wrote automated tests for instruction execution, flag behavior, and memory I/O.

CPU Simulator - YESS Y86 - C++

- Developed a Y86 simulator in C++ supporting ASCII object file parsing and execution.
- Created a custom parser and debugger for validating object code at each execution step.
- Achieved accurate simulation of the Y86 instruction set through rigorous validation and test coverage.
- Ensured full compatibility with the Y86 instruction format and met all project specs.

Esoteric (BF) Language Interpreter – Java

- Engineered a minimalistic interpreter for the BF language in Java, showcasing mastery of pointer-based memory models and low-level control logic.
- Implemented core features such as dynamic memory cells, character-level input/output, and loop-based execution flow.
- Strengthened debugging and algorithmic reasoning by managing edge cases, nested loops, and invalid syntax gracefully.

TECHNICAL SKILLS

Languages: Python, C#, Java, C/C++, Zig, SQL, HTML/CSS, JavaScript, R, Haskell, OCaml, Bash

Frameworks: React, FastAPI, REST APIs, Node.js

Tools: Git, Linux, Maven